MAJOR PROJECT STAGE 1

GENAI-ASSISTED AML COMPLIANCE SYSTEM

REVIEW 3

SUBMITTED BY

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1. ABSTRACT

Money laundering is a major financial crime that threatens the integrity of financial institutions and the global economy. Conventional Anti-Money Laundering (AML) systems mainly rely on predefined rules, fixed thresholds, and manual investigation. These approaches often generate excessive false positives, require considerable human effort, and may fail to detect complex or previously unseen laundering patterns.

This project proposes a GenAI-Assisted AML Compliance System that combines Machine Learning, Artificial Intelligence, Natural Language Processing, Generative AI, and explainable analytics to strengthen transaction monitoring and suspicious activity detection. The system analyzes transaction amount, frequency, location, account behavior, beneficiary relationships, and transaction history to identify anomalous patterns and assign customer or transaction risk scores. Generative AI assists compliance analysts by providing understandable reasons for alerts, summarizing investigation findings, identifying contributing risk factors, and supporting the preparation of compliance reports.

A secure investigation dashboard enables authorized personnel to review alerts, examine customer and transaction risk information, manage investigation status, and prioritize high-risk cases. The system also incorporates access controls, audit logging, data protection, explainability, and human validation to support responsible AI use and regulatory compliance.

By reducing unnecessary alerts and improving investigation efficiency, the proposed solution provides a scalable and interpretable decision-support system while ensuring that final compliance decisions remain under human oversight.

Keywords: Anti-Money Laundering, Transaction Monitoring, Anomaly Detection, Risk Scoring, Generative AI, Explainable AI, Suspicious Activity Report, Compliance Dashboard.

2. INTRODUCTION

2.1 Background

Money laundering is the process of disguising the proceeds of crime so that they appear to originate from legitimate sources. It typically passes through three stages: placement (introducing illicit cash into the financial system), layering (moving funds through many accounts and transactions to obscure their origin) and integration (re-introducing the funds into the economy as apparently legitimate wealth). Banks, payment providers and other regulated entities are legally required to monitor customer activity and report suspicious transactions to the financial intelligence authority of their country, in line with international standards such as the FATF Recommendations. In India, these obligations arise mainly from the Prevention of

Money Laundering Act (PMLA), 2002 and the guidelines issued by the Reserve Bank of India and the Financial Intelligence Unit–India (FIU-IND).

Most institutions still depend on rule-based monitoring, in which a transaction is flagged when it crosses a fixed threshold or matches a predefined scenario. Because criminals continuously adapt their behaviour to stay below such thresholds, rule-based systems produce large numbers of false alerts while still missing new typologies. Analysts then spend most of their time reviewing harmless alerts and writing narrative reports manually.

2.2 Motivation

Recent progress in machine learning allows behaviour-based detection of unusual activity, and advances in Generative AI and Natural Language Processing make it possible to explain model outputs and draft investigation narratives in plain language. However, a risk score alone is not sufficient in a regulated environment: analysts and auditors must understand why an alert was raised and must remain accountable for the final decision. This project is motivated by the need for an AML solution that is accurate, explainable, auditable and designed around human oversight.

2.3 Objectives

- To design a transaction-monitoring system that detects anomalous and potentially suspicious activity using machine learning.
- To assign interpretable risk scores to customers and transactions using amount, frequency, location, account behaviour, beneficiary relationships and transaction history.
- To reduce false-positive alerts and help analysts prioritise high-risk cases.
- To use Generative AI for alert explanations, investigation summaries, identification of risk factors and drafting of compliance reports.
- To provide a secure investigation dashboard with role-based access, case management and audit logging.
- To keep a human analyst in the loop so that every final compliance decision is validated by an authorised person.

2.4 Scope

The system is a decision-support tool for AML compliance teams. It covers data ingestion, feature engineering, anomaly detection, risk scoring, explainability, GenAI-assisted summarisation and report drafting, and an investigation dashboard. The prototype is developed and evaluated on synthetic or anonymised transaction data; it does not automatically block

accounts or file reports with regulators without human approval. Integration with live core-banking systems is considered future work.

3. LITERATURE REVIEW

Research on AML technology has moved from fixed rules to statistical learning, graph analysis, explainable models and, most recently, large language models. The table below summarises the main directions that inform this project.

Approach	Representative work	Strengths	Limitations
Rule / threshold-based monitoring	FATF Recommendations; typology-based scenario rules used in commercial AML tools	Simple, transparent and easy to justify to regulators	High false-positive rate; static; misses new patterns; needs frequent manual tuning
Supervised machine learning	Gradient boosting (Chen & Guestrin, XGBoost, 2016); Random Forest	Strong accuracy when labelled data is available	Confirmed laundering labels are rare and imbalanced; poor on unseen typologies
Unsupervised anomaly detection	Isolation Forest (Liu, Ting & Zhou, 2008); autoencoders	No labels needed; finds unusual behaviour	Anomalous does not always mean suspicious; scores are hard to explain
Graph-based learning	Weber et al., "Anti-Money Laundering in Bitcoin" using Graph Convolutional Networks (2019); Altman et al., synthetic AML transaction datasets (NeurIPS 2023)	Captures networks of accounts and layering patterns	Computationally heavy; needs relationship data; limited interpretability
Explainable AI	SHAP (Lundberg & Lee, 2017); LIME	Shows which features drove each score; supports audit	Outputs are technical and not directly readable by non-technical staff
Large language models / GenAI	Large pre-trained language models (e.g., Brown et al., 2020)	Produces fluent explanations, summaries and report drafts	Can hallucinate; needs grounding in verified data and human review

3.1 Research Gap

Existing literature treats detection, explanation and reporting largely as separate problems. Machine-learning studies focus on improving detection metrics but give analysts little help in understanding alerts, while language-model studies seldom connect to a real alert and case-management workflow with access control and audit trails. There is therefore a need for an integrated system in which (i) a machine-learning engine detects and scores risk, (ii) explainability methods identify the contributing factors, (iii) Generative AI converts those

factors into understandable explanations and report drafts grounded in the underlying data, and (iv) humans validate every final decision. This project addresses that gap.

Note: *references listed above should be verified and formatted in the department-prescribed style (IEEE/APA) in the final report.*

4. PROBLEM STATEMENT

Financial institutions process millions of transactions every day, and regulators expect suspicious activity to be identified and reported promptly. Traditional AML monitoring based on fixed rules and thresholds creates too many false positives, depends heavily on manual investigation, and cannot reliably detect complex or previously unseen laundering patterns. Even where machine-learning models are used, their outputs are often opaque, and analysts must still interpret scores and write case narratives by hand.

4.1 Key Problems Identified

- **Excessive false positives:** the large majority of alerts generated by rule-based systems are not genuine suspicious cases, wasting investigator time.
- **Manual effort:** reviewing alerts, collecting customer and transaction history, and writing compliance reports are slow, repetitive tasks.
- **Missed complex patterns:** structuring, layering through multiple accounts, and new typologies are difficult to capture with static rules.
- **Lack of explainability:** analysts and auditors cannot easily see why a customer or transaction was flagged.
- **Weak prioritisation:** alerts are often handled in arrival order instead of by risk, delaying high-risk cases.
- **Governance and security concerns:** sensitive financial data requires access control, audit logging, and responsible use of AI with human oversight.

4.2 Problem Definition

To design and develop an explainable, secure and scalable decision-support system that analyses transaction amount, frequency, location, account behaviour, beneficiary relationships and transaction history to detect anomalous activity, assign risk scores, reduce unnecessary alerts, and use Generative AI to explain alerts and assist report preparation, while keeping final compliance decisions under human control.

5. EXISTING SYSTEM

In most banks and financial institutions today, AML compliance is performed by a transaction-monitoring system built around predefined rules and fixed thresholds, followed by manual investigation by compliance analysts.

5.1 Working of the Existing System

1. Transactions are collected from core banking and payment channels.
2. Static rules are applied, for example: cash deposits above a set limit, many transactions just under a reporting threshold, rapid in-and-out movement of funds, or transfers to high-risk jurisdictions.
3. Every transaction matching a rule generates an alert, regardless of customer context.
4. Analysts manually open each alert, gather customer profile and past transaction history from several systems, and judge whether it is suspicious.
5. For suspicious cases, the analyst writes the narrative manually and prepares a Suspicious Transaction/Activity Report for the compliance officer.

5.2 Limitations of the Existing System

- Fixed thresholds can be learnt and avoided by criminals, and are not adapted to individual customer behaviour.
- A very high proportion of alerts are false positives, causing alert fatigue and backlog.
- Unable to detect new or complex laundering patterns that do not match any written rule.
- Little or no explanation of why an alert was raised beyond the name of the rule.
- Information is scattered across systems, making investigation slow and inconsistent.
- Report writing is manual, time-consuming and varies from analyst to analyst.
- Rule maintenance requires continuous manual tuning and does not scale with growing transaction volumes.

6. PROPOSED SYSTEM

6.1 Overview

The proposed GenAI-Assisted AML Compliance System combines machine learning, explainable analytics and Generative AI in a single workflow. Transaction and customer data are cleaned and converted into behavioural features; machine-learning models detect anomalies and assign risk scores; explainability techniques identify the factors responsible; and a Generative AI component turns these factors into clear explanations, case summaries and draft

compliance reports. Authorised analysts review everything on a secure dashboard and make the final decision.

6.2 Key Features

- **Behaviour-based detection:** analyses amount, frequency, location, account behaviour, beneficiary relationships and transaction history instead of only fixed thresholds.
- **Customer and transaction risk scoring:** every transaction and customer receives a risk score so that high-risk cases are prioritised.
- **Explainable alerts:** contributing risk factors are listed for each alert (for example, using SHAP values).
- **GenAI assistance:** plain-language alert reasons, investigation summaries and drafts for compliance reports, grounded only in the case data.
- **Secure investigation dashboard:** alert queue, customer and transaction risk views, case status management and priority ranking.
- **Responsible AI controls:** role-based access control, audit logging, data protection, and mandatory human validation of decisions.

6.3 Workflow of the Proposed System

1. Ingest transaction, customer and beneficiary data and validate it.
2. Pre-process the data and engineer behavioural features (e.g., transaction velocity, deviation from customer average, new-beneficiary flag, geographic risk).
3. Run the machine-learning anomaly-detection model and compute risk scores.
4. Generate alerts for cases above the risk threshold and compute contributing factors.
5. Use Generative AI to write the alert reason and case summary from the verified factors.
6. Analyst reviews the alert on the dashboard, adds notes and updates the investigation status.
7. If suspicious, a draft compliance report is generated, edited and approved by the compliance officer.
8. All actions are written to the audit log; analyst feedback is used to improve the model and reduce false positives.

6.4 Comparison: Existing vs Proposed System

Aspect	Existing System	Proposed System
Detection method	Fixed rules and thresholds	Machine-learning behaviour analysis plus risk scoring

Aspect	Existing System	Proposed System
False positives	High	Reduced through risk-based alerting and feedback
New patterns	Mostly missed	Anomaly detection can flag unseen patterns
Explainability	Only rule name	Contributing factors and GenAI plain-language reasons
Reporting	Fully manual	GenAI-drafted summaries and reports, human-approved
Prioritisation	Arrival order	Ranked by risk score
Governance	Limited traceability	RBAC, audit logs and human-in-the-loop validation

6.5 Advantages

- Fewer unnecessary alerts and faster investigations.
- Better detection of complex and evolving laundering patterns.
- Transparent, interpretable and auditable decision support.
- Scalable to large transaction volumes.
- Final decisions remain with human analysts, supporting regulatory acceptance.

7. HARDWARE REQUIREMENTS

The following hardware is recommended for developing, training and running the prototype. Production deployment would be sized according to transaction volume.

7.1 Development / Client System

Component	Minimum / Recommended Specification
Processor	Intel Core i5 (8th gen) or AMD Ryzen 5 or higher
RAM	8 GB minimum; 16 GB recommended for model training
Storage	256 GB SSD minimum; 512 GB SSD recommended
GPU (optional)	NVIDIA GPU with 4 GB+ VRAM for faster model training / local LLM inference
Display	1366 x 768 minimum; 1920 x 1080 recommended
Network	Broadband internet connection (for package installation and LLM API access)
Input devices	Standard keyboard and mouse

7.2 Server / Deployment System (Prototype)

Component	Specification
Processor	4 vCPUs or more (Intel Xeon / AMD EPYC class)
RAM	16 GB or more
Storage	500 GB SSD with regular backup
GPU (optional)	NVIDIA T4 class or better if a local open-source LLM is hosted
Network	Secure, encrypted connectivity (HTTPS/TLS) with firewall protection

8. SOFTWARE REQUIREMENTS

The software stack is based on open-source tools that are widely used for data science and web application development.

Category	Software / Technology
Operating system	Windows 10/11, Ubuntu 22.04 LTS or macOS
Programming language	Python 3.10+ (backend and ML); JavaScript/TypeScript (front end)
Data processing	Pandas, NumPy
Machine learning	Scikit-learn, XGBoost / LightGBM, Isolation Forest, imbalanced-learn
Explainability	SHAP (and optionally LIME)
NLP / Generative AI	Large Language Model via API or locally hosted open-source model; LangChain / prompt templates; Hugging Face Transformers
Backend framework	FastAPI or Flask (REST API with JWT-based authentication)
Front end / dashboard	React.js (or Streamlit for prototype), Plotly / Chart.js for visualisation
Database	PostgreSQL (or MySQL); optional vector store for case notes
Security	JWT / OAuth2, role-based access control, HTTPS/TLS, password hashing, encrypted storage
Development tools	VS Code, Jupyter Notebook / Google Colab, Git and GitHub, Postman
Datasets	Synthetic / anonymised transaction datasets (e.g., IBM synthetic AML data, PaySim)
Browser	Google Chrome, Microsoft Edge or Firefox (latest versions)

9. SYSTEM ARCHITECTURE – DESCRIPTION

The system follows a layered architecture so that each concern (data, analytics, GenAI, application and security) can be developed, tested and scaled independently. Figure 9.1 shows the overall architecture.

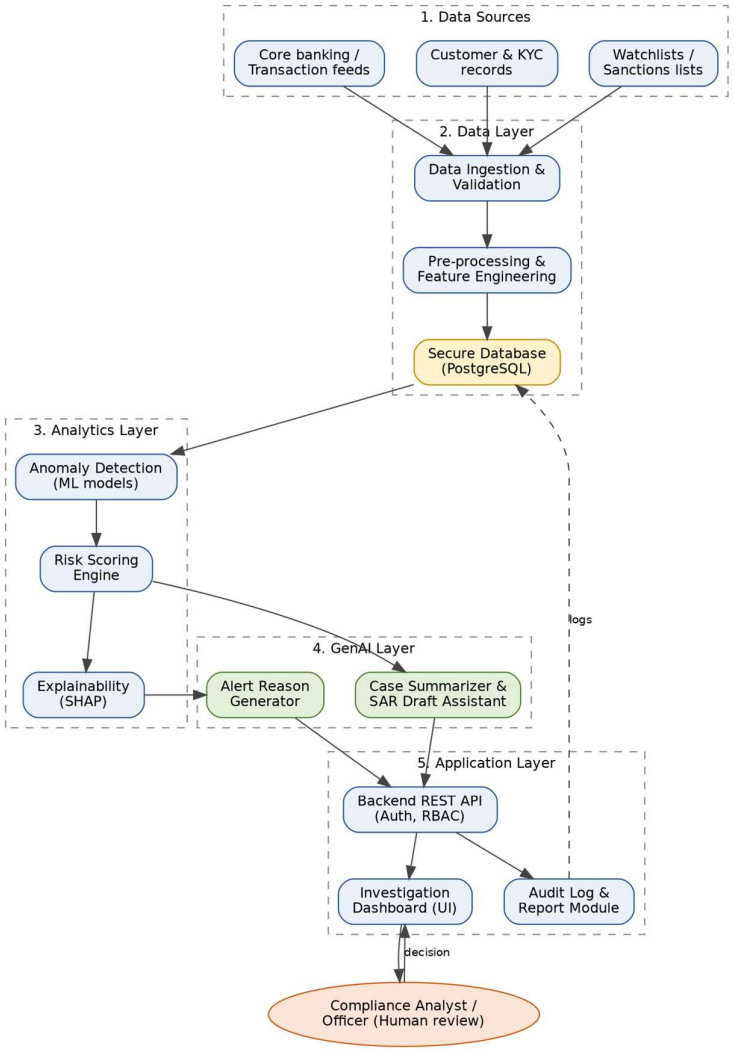


Figure 9.1: System Architecture of the GenAI-Assisted AML Compliance System

9.1 Description of Layers

- **Data sources:** transaction feeds from banking/payment systems, customer and KYC records, and sanction/watchlist data are the inputs to the system.
- **Data layer:** the ingestion component validates and cleans incoming data; the pre-processing and feature-engineering component derives behavioural features such as transaction velocity, deviation from the customer's normal amount, geographic risk and beneficiary relationship indicators. Results are stored in a secure database.

- **Analytics layer:** the anomaly-detection models identify unusual behaviour; the risk-scoring engine combines model output with customer risk attributes to produce transaction and customer risk scores; the explainability component computes the contribution of each feature to the score.
- **GenAI layer:** the alert-reason generator converts the verified risk factors into plain-language explanations, and the case summariser and report assistant produces investigation summaries and draft compliance reports. The language model receives only structured, verified case data to limit hallucination.
- **Application layer:** a backend REST API handles authentication, role-based access and business logic; the investigation dashboard presents alerts, risk information and case status; the audit and report module records every action and stores approved reports.
- **Human oversight:** the compliance analyst or officer reviews the AI output, makes the final decision, and the feedback is returned to improve the system.

9.2 Data Flow

Raw data enters through the ingestion component, passes through validation and feature engineering, and is stored in the database. The analytics layer reads the features, produces risk scores and explanations, and creates alerts. The GenAI layer enriches each alert with a readable explanation and summary. The API serves these to the dashboard, where the analyst records a decision; the decision and all user actions are saved back to the database and audit log.

9.3 Security and Responsible-AI Design

- Role-based access control separating analysts, compliance officers and administrators.
- Encryption of data in transit (TLS) and at rest; masking of sensitive customer fields before they are sent to the language model.
- Immutable audit logging of logins, views, status changes and report approvals.
- Human-in-the-loop validation: the AI never closes a case or files a report on its own.
- Model and prompt versioning to allow reproducibility and review.

10. MODULES – DESCRIPTION

The system is divided into the following eight modules.

10.1 Data Ingestion and Validation Module

Attribute	Description
Purpose	To collect and validate transaction, customer and beneficiary data.
Input	Transaction records (CSV/database/API), customer and KYC data, watchlists.

Attribute	Description
Processing	Schema and type checks, removal of duplicates, handling of missing values, normalisation of currency and timestamps.
Output	Clean, validated records stored in the secure database.

10.2 Pre-processing and Feature Engineering Module

Attribute	Description
Purpose	To convert raw data into behavioural features that describe how a customer transacts.
Input	Validated transaction and account data.
Processing	Computes transaction amount statistics, frequency/velocity over time windows, deviation from customer baseline, location and geography risk, new or unusual beneficiary flags, and rapid in-and-out fund movement; encodes and scales features.
Output	Feature set ready for model training and scoring.

10.3 Anomaly Detection and Risk Scoring Module

Attribute	Description
Purpose	To identify anomalous or suspicious patterns and assign risk scores.
Input	Engineered features.
Processing	Applies unsupervised (e.g., Isolation Forest) and supervised (e.g., XGBoost) models, handles class imbalance, and combines model output with customer risk attributes into a normalised risk score and risk category (Low/Medium/High).
Output	Transaction and customer risk scores; alerts for scores above the threshold.

10.4 Explainability Module

Attribute	Description
Purpose	To make every score understandable and auditable.
Input	Model, features and risk score of an alert.
Processing	Computes feature contributions using SHAP and ranks the top contributing risk factors.
Output	Ordered list of risk factors for each alert.

10.5 GenAI Assistant Module

Attribute	Description
Purpose	To help analysts understand alerts and prepare documents faster.
Input	Alert details, contributing risk factors, customer and transaction history (masked).
Processing	Uses structured prompt templates with the verified data to generate plain-language alert reasons, investigation summaries and draft compliance report narratives; includes guardrails against unsupported statements.
Output	Alert explanation, case summary and draft report text for analyst review.

10.6 Investigation Dashboard Module

Attribute	Description
Purpose	To give authorised users a single interface for alert review and case management.
Input	Alerts, risk scores, explanations and summaries.
Processing	Displays prioritised alert queue, customer and transaction risk views, charts and filters; lets analysts add notes, assign cases and update investigation status.
Output	Updated case status, analyst notes and decisions.

10.7 Security, Access Control and Audit Module

Attribute	Description
Purpose	To protect data and ensure accountability.
Input	User credentials and all user actions.
Processing	Authenticates users, enforces role-based permissions, encrypts sensitive data and records an audit trail of every important action.
Output	Secure sessions and a tamper-evident audit log.

10.8 Reporting Module

Attribute	Description
Purpose	To prepare and store compliance reports.
Input	Approved case information and GenAI draft narrative.
Processing	Fills a report template, allows the analyst to edit, routes to the compliance officer for approval and stores the final version.
Output	Approved suspicious activity report (exportable as PDF/Word).

11. UML DIAGRAMS

The following UML diagrams describe the functional and structural design of the system.

11.1 Use Case Diagram

The use case diagram shows the three actors – Compliance Analyst, Compliance Officer and System Administrator – and the functions available to each. Analysts review alerts and prepare reports, officers approve reports and view audit logs, and administrators manage users and data.

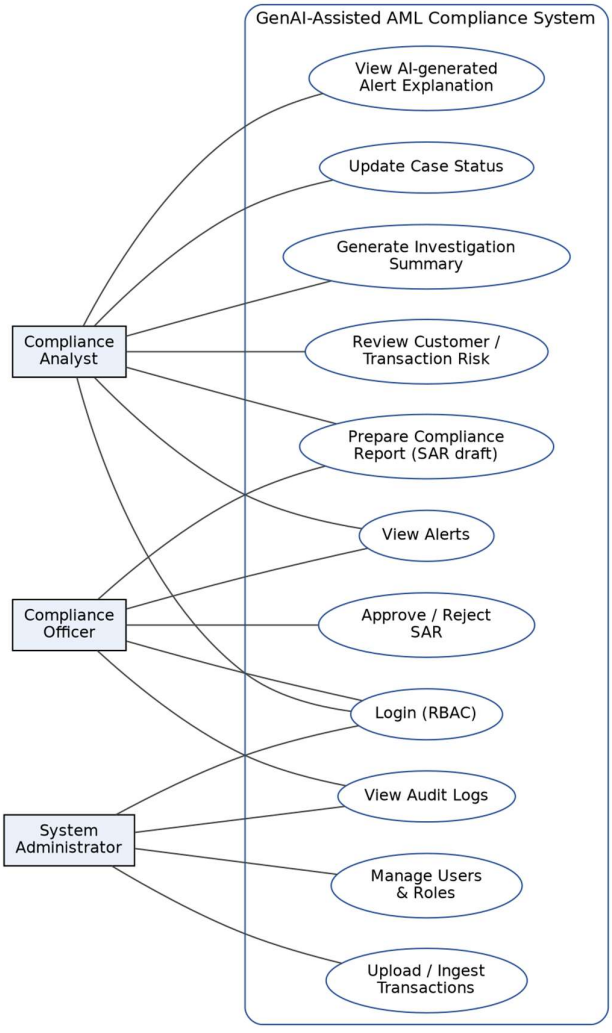


Figure 11.1: Use Case Diagram

11.2 Class Diagram

The class diagram presents the main entities: a Customer owns Accounts, Accounts contain Transactions, and Transactions are directed to Beneficiaries. The AnomalyModel scores transactions and generates Alerts; an Alert opens an InvestigationCase, which may produce a ComplianceReport. The GenAIService explains alerts and summarises cases, and User actions are recorded in the AuditLog.

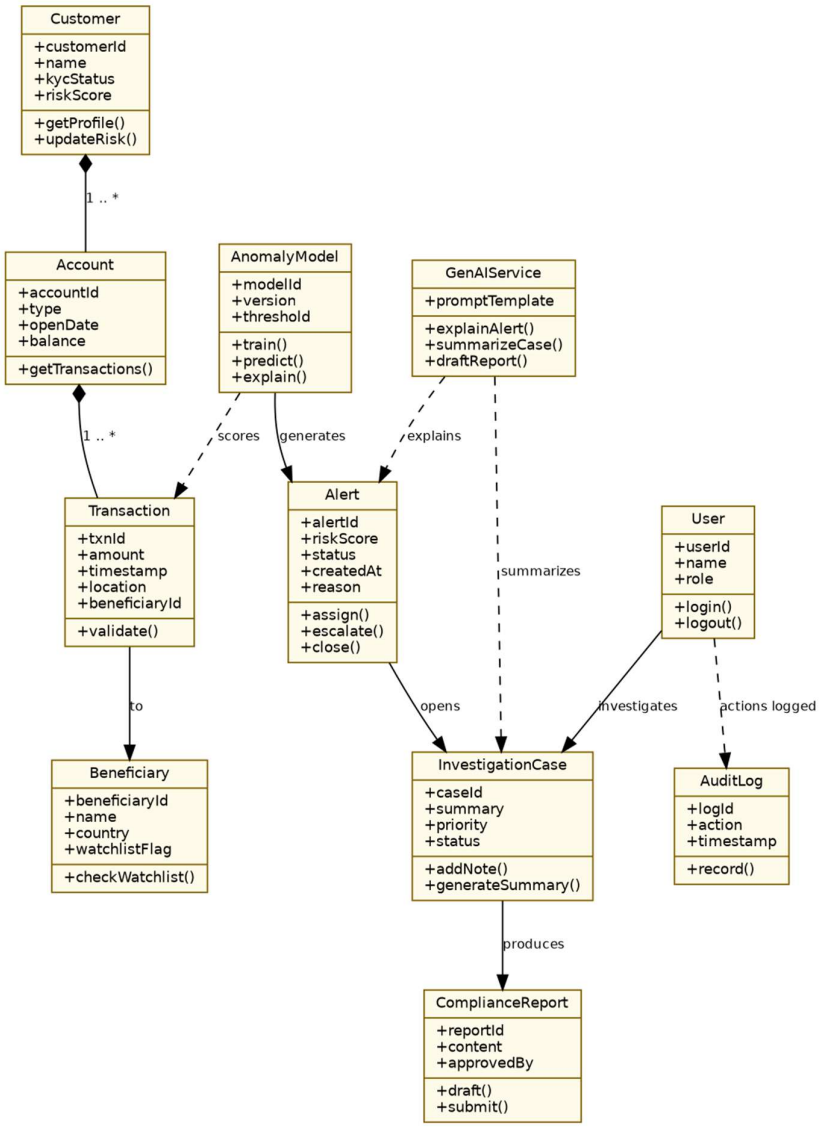


Figure 11.2: Class Diagram

11.3 Sequence Diagram

The sequence diagram shows how an analyst reviews an alert: the dashboard requests alert data from the backend, which retrieves scores from the database, obtains contributing factors from the risk engine and an explanation from the GenAI service; the analyst then records a decision, which is stored with an audit entry.

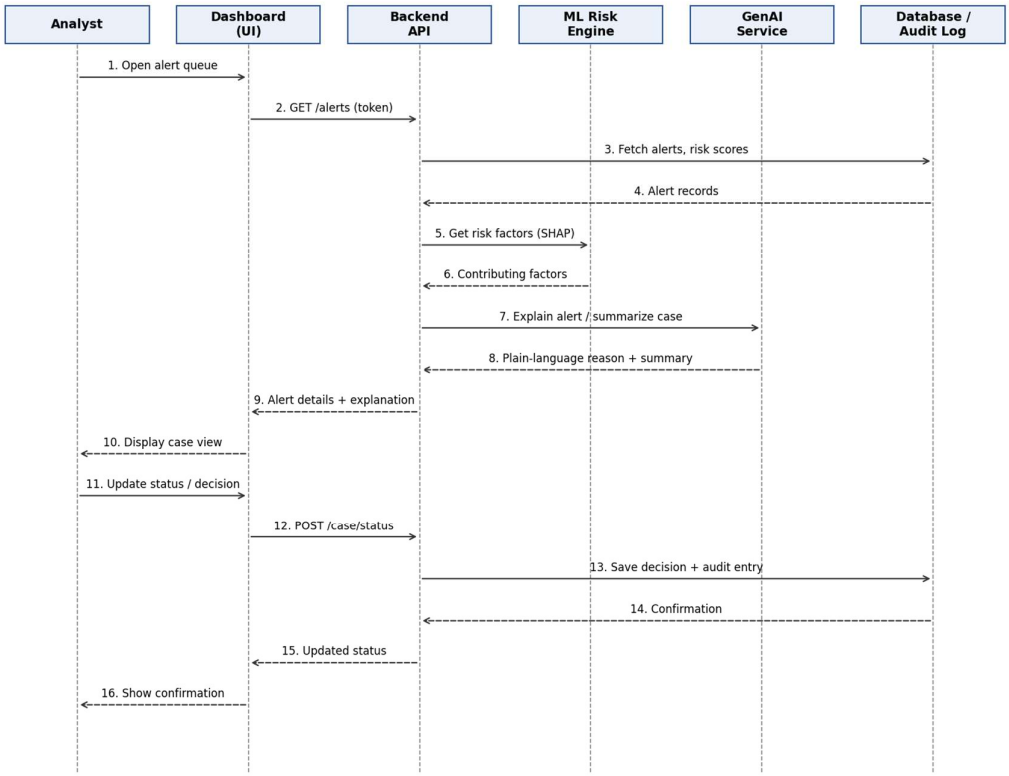


Figure 11.3: Sequence Diagram – Alert Review and Decision

11.4 Activity Diagram

The activity diagram shows the end-to-end flow from data ingestion to risk scoring, alert generation, GenAI-assisted review, the analyst's decision, reporting and audit logging.

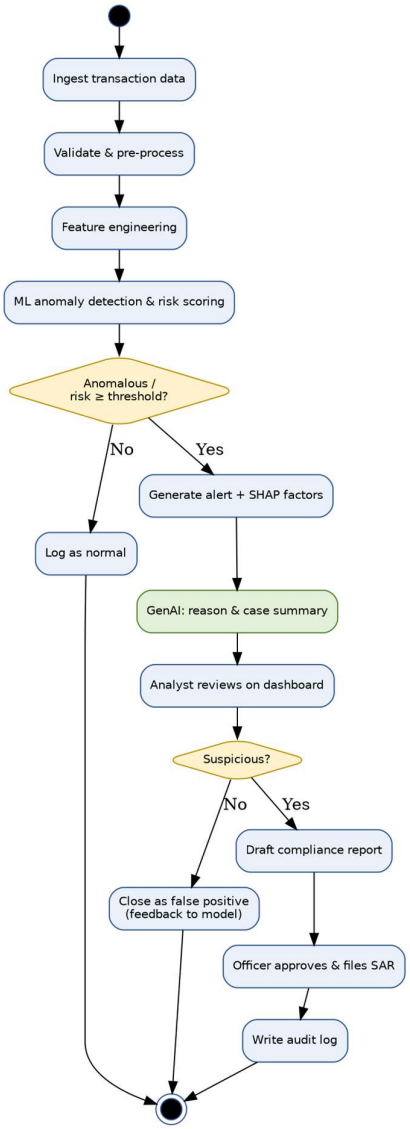


Figure 11.4: Activity Diagram – AML Alert Processing